Chapter 9, Problem 8PP

Problem

If voltage $v = 25\sin(100t - 15°)$ V is applied to a 50 μF capacitor, calculate the current through the capacitor.

Step-by-step solution

Step 1 of 3

Write the given sinusoidal expression of voltage.

$$v = 25\sin(100t - 15°)\ \text{V}$$

Here,

The magnitude of the voltage v_m is 25 V

The angular frequency ω is 100 rad/s

The phase angle is $-15°$.

The phasor representation of voltage is,

$$\mathbf{V} = 25\angle -15°\ \text{V}$$

Step 2 of 3

Determine the reactance of $50\ \mu\text{F}$ capacitor.

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(100)(50\times10^{-6})} \\ &= -j200 \\ &= 200\angle -90°\ \Omega \end{aligned}$$

Step 3 of 3

Calculate the current through the capacitor.

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}}{X_C} \\ &= \frac{25\angle -15°}{200\angle -90°} \\ &= 0.125\angle 75°\ \text{A} \\ &= 125\angle 75°\ \text{mA} \end{aligned}$$

Convert the current expression from phasor domain to time domain.

$$\begin{aligned} i &= i_m \sin(\omega t + \phi) \\ &= 125\sin(100t + 75°)\ \text{mA} \end{aligned}$$

Therefore, the current through the capacitor is $\boxed{125\sin(100t + 75°)\ \text{mA}}$.